

Systems that refuse to invent.

Kunal Kumar

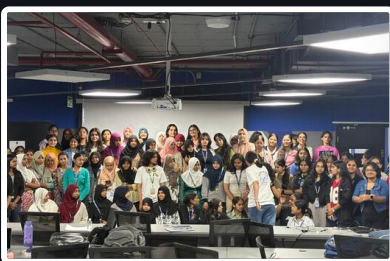
Computer Science, class of 2028 · Bengaluru, India
Frontend Engineering Intern, Emergent (YC S24)

MATS round 2

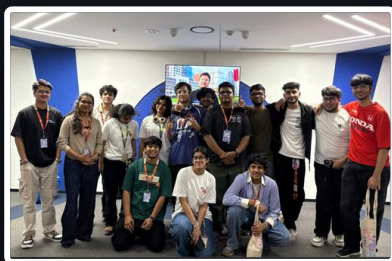
Activate AI Fellow · 1 of 15

HPAIR 2026, Harvard

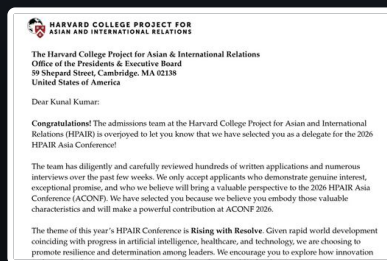
1,500 students reached



1,500 students, one room at a time



The crew behind the events



Selected delegate, HPAIR 2026

Every system I use will guess, then deliver the guess in the same voice it uses for facts. **An honest “I don't know” is harder to build than a confident answer**, and nobody is rewarded for building it.

So that is what I build. **Five projects, one idea**, arrived at five separate times before I noticed it had become a thesis.

32 repositories · 829 contributions in twelve months · 29 patches merged into 35 open-source projects.

01

Recall

A memory engine for unfinished work

02

ATTEST

Money that moves only on proof

03

Support Triage

An agent that cannot be talked into it

04

Minutes

A meeting workspace, rebuilt

05

PDFChat

Cites its pages, or refuses

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Recall

0

MODELS IN THE PRODUCTION PATH

A memory engine for unfinished work. One keystroke brings back the tabs, files and chats from your last session.

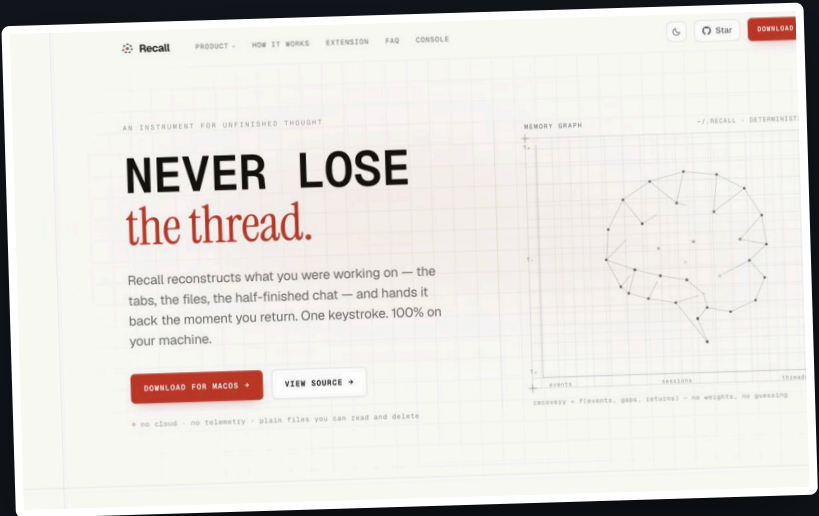
Someone interrupts you at 3pm. You sit back down and everything is still there, and none of it helps. **You did not lose files. You lost continuity.**

Spotlight and grep assume you remember a string. A cognitive restart asks *what was I doing*, and nothing on the machine answers that.

TECHNICAL ASPECTS

Seven layers over a local event log. Press Ctrl + Space and the moment, the session and the topic all come back at once.

No model sits in the production path, so the same question can never return two different pasts. Everything it knows lives in ~/ .recall/ as plain JSON you can read with cat.



YEAR	2026
ROLE	Sole author
BUILT WITH	Python · PyQt6 · FastAPI · ChromaDB · Tauri
LINK	web-seven-puce-99.vercel.app github.com/kunalKumar-13/Recall-me

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ATTEST

₹0.00

MOVED AGAINST THE WRONG ACCOUNT

A financial control layer. It verifies what an AI proposes before anything is allowed to move money.

A payment gateway sends one lump sum covering many orders, and **more than one set of orders can add up to the same credit exactly.**

Those sets discharge *different customers*. Guessing marks the wrong one as paid while the books still balance, so nobody finds out until it is expensive.

TECHNICAL ASPECTS

So ATTEST refuses. Across 500 settlements it had never seen, **82.4% were correctly refused.**

An **advisor** proposes which records to look at and is **never given an amount**. A **verifier** sharing no code with it re-derives every explanation. The benchmark names a competitor that beats mine on precision, because a comparison curated in your own favour proves nothing.



YEAR	2026
ROLE	Sole author
BUILT WITH	Python · Rust solver kernel
LINK	github.com/kunalKumar-13/attest

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Support Triage

132/132

EVALUATION TESTS PASSING

A support agent where code owns the control flow and the model is a constrained analyst, not the decision-maker.

An agent that can be talked into a destructive action is worse than no agent. **Its worst possible behaviour here is: escalate to a human.**

The usual design hands the model the wheel and hopes the prompt holds.

TECHNICAL ASPECTS

The LLM is invoked once per ticket, inside an unforgeable sentinel block, and returns typed JSON a policy validator adjudicates. It cannot decide escalation, execute tools, or see raw PII.

Defends prompt injection in seven languages. When retrieved chunks contradict each other, confidence drops and the validator escalates rather than picking a side.

```
$ triage --ticket 4417

  ingest    sealed in <<<TICKET>>>
 retrieve   4 chunks · 2 disagree
 analyse    LLM called once, in sentinel block
 policy     confidence 0.41 · cap 0.60

DECISION    ESCALATE TO HUMAN

$ sha256sum run_a.csv run_b.csv
2f9c...a71b run_a.csv
2f9c...a71b run_b.csv
```

YEAR	2026
ROLE	Sole author
BUILT WITH	Python · LlamaIndex · BM25 · TF-IDF
LINK	github.com/kunalKumar-13/ deterministic-support-triage

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Minutes

2

COMMANDS TO A POPULATED PRODUCT

A meeting workspace with interactive transcripts, AI notes and workspace-wide search, rebuilt from the product outward.

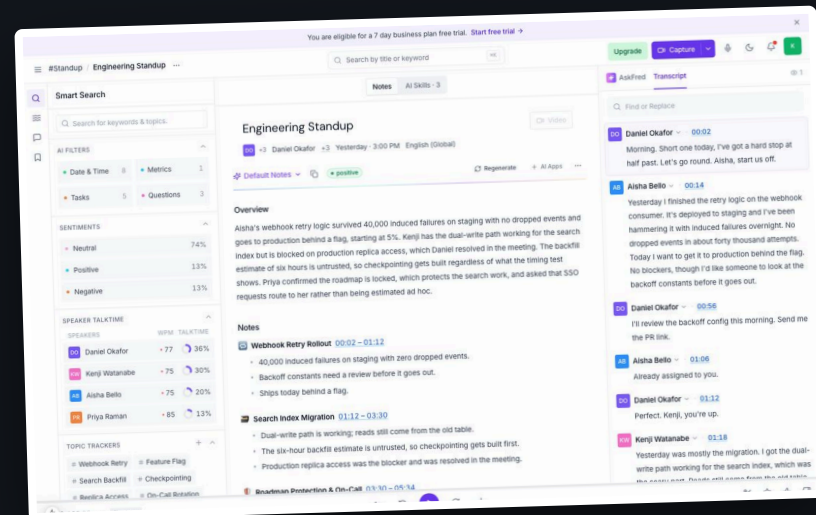
A meeting product is not its landing page. **Everything valuable happens after the meeting ends.**

“Clone this product” is the easiest brief in the world to do badly: rebuild the hero section, ship something that collapses when anyone clicks.

TECHNICAL ASPECTS

So I set a bar I could not fake: everything except the speech-to-text actually works, and **the README says so on the first screen**, before a reviewer can find out.

Next.js 15 over FastAPI, with **SQLite FTS5** as a real index. The decision that mattered: I stopped matching screens and **extracted the token system instead**, after which twenty-plus screens came out coherent untouched.



YEAR	2026
ROLE	Sole author · assignment brief
BUILT WITH	Next.js 15 · React 19 · TypeScript · FastAPI
LINK	github.com/kunalKumar-13/minutes

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PDFChat

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UNCITED ANSWERS

Ask a document anything. Every answer cites its pages, or says plainly that the document does not support the question.

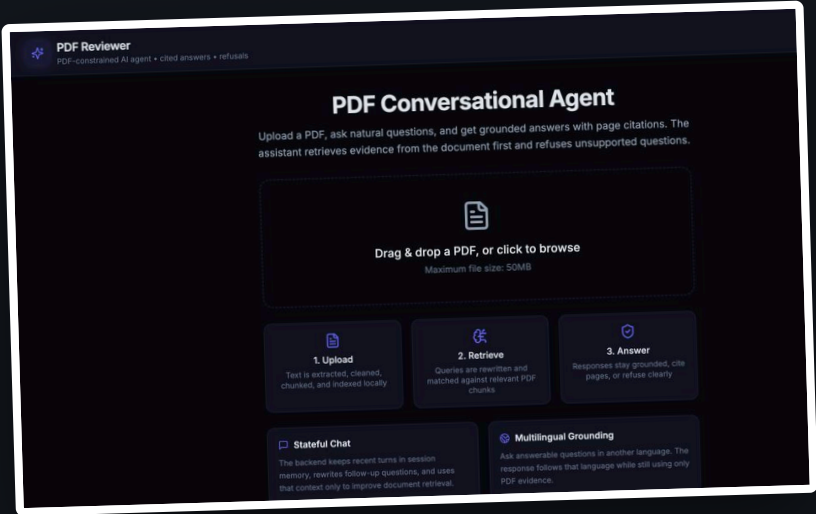
A confident wrong answer costs more than no answer. **Every response either cites the pages it came from, or refuses.**

Retrieval gets messy, and the standard failure is a model that fills the gap fluently and invisibly.

TECHNICAL ASPECTS

Extraction, chunking and indexing run locally, and queries are rewritten against the relevant chunks before generation.

The refusal path is a first-class branch, not an error handler. If retrieval does not support an answer, that *is* the answer. It answers in your language while still treating the source as the only truth.



YEAR	2026
ROLE	Sole author
BUILT WITH	Python · PyMuPDF · TF-IDF · FastAPI
LINK	pdf-reviewer-ten.vercel.app github.com/kunalKumar-13/PDF-Reviewer

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The rest of me

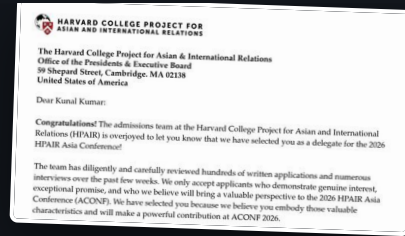
Research, rooms I did not belong in, people who owed me nothing, and 1,500 students.



What I am chasing

Three projects here turned out to be one question: when a model proposes something, **what is structurally permitted to believe it?**

I read alignment work the way other people read the news.



Rooms I do not belong in

Selected delegate to **HPAIR 2026 at Harvard**, arguing about things I had no technical claim on.

Every project here began as a problem I could name in one sentence. **That habit came from that room, not a codebase.**



Strangers grade my code

OpenStreetMap's editor, Django's own site, a Bitcoin Lightning wallet, MetaBrainz, Plone, the **Fortran standard library**.

I had no reason to know Fortran. **The skill is arriving somewhere I have no standing and becoming useful.**



I make things happen for other people

Growth & Ops Lead at **Hack Club**, 16 months, reaching **1,500 students**. Plus 50+ mentored in DSA.

At that scale you stop preventing failure and design the degradation. **Every system here has a documented failure mode.**

Why this Academy

Everything I build lives on a screen someone is already sitting at. I have never built for the device someone holds *while something is happening to them*, and that is a different discipline, not the same one scaled down. Three times now the hardest part of an engineering project turned out to be a design problem I had no vocabulary for. **An honest "I don't know" is worth nothing if nobody can read it.** That is the half I am asking you to teach me.

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